Spin-channel redistribution in quadrupole shape diagnostics of odd barium

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Abstract

Convergence of a quadrupole diagnostic within one intermediate-spin channel does not establish how its evolution relates to intrinsic charge shape. An unpaired particle–rotor control isolates this question by fixing the charge tensor at $\gamma = \mathbf{20}^\circ$: complete moments remain constant while the fully summed selected-channel ratio changes and leaves the real-angle domain. We then resolve the coupling response in the lowest $\mathbf{11/2}^-$ states of $^{\mathbf{129-137}}$Ba within an explicitly specified interacting boson–fermion model. Five states exhaust more than **99.5%** of the selected-channel quadratic strength but only **20.0**–**28.9%** of the complete moment. A fixed-operator intervention identifies the redistribution of static strength and signed cubic paths behind their different responses. Alternative charge calibrations and a failed conditional high-spin isomer check limit the empirical interpretation of the Ba calculation. The fixed-geometry counterexample and the finite-model mechanism together show why angular-channel coverage and coupling response must be assessed separately from state-count convergence when converting a selected diagnostic into a shape coordinate.

Keywords: quadrupole invariants, odd-mass nuclei, interacting boson–fermion model, barium isotopes

1 Introduction

Electric-quadrupole matrix elements connect nuclear spectroscopy to collective shape. Their scalar products describe charge distributions without choosing a laboratory orientation, and their relation to deformation has long been developed in the quadrupole-invariant formalism [1]. In practice, only a restricted matrix-element network may be available. The quality of a deformation estimate then depends on which matrix elements are retained as well as how accurately they are known. Approximations using a few observables and the convergence of finite sums have been examined in rotor, interacting-boson and shell-model calculations [2, 3].

This distinction is particularly relevant to odd nuclei. A rank-two tensor acting on a state with spin J connects several intermediate spins, and particle–core coupling can redistribute strength among them. Nomura and collaborators already showed that indicators based on different ΔJ selections can evolve differently in odd-mass Eu and Sm [4]. For the γ-soft region near $A = 130$, their interacting boson–fermion model (IBFM) study used effective quadrupole coordinates constructed from the lowest five $J+2$ states [5]. Such a selection defines legitimate rotational scalars. Its convergence, however, does not establish equivalence to the complete charge-polynomial moments.

We test this inference with a rigid-charge particle–rotor control that holds the intrinsic tensor exactly fixed while varying angular coupling. A separate odd-Ba IBFM calculation then compares complete and restricted quantities on

the same states, at fixed initial spin and with the same electric-quadrupole operator. Its controlled coupling path resolves the redistribution of static strength and signed cubic contributions. Electromagnetic constraints delimit the application of this mechanism to the actual nuclei. The Ba study uses published Hamiltonian parameters but does not reproduce or correct the published electromagnetic-invariant figures.

2 Model and quadrupole diagnostics

2.1 Fixed Hamiltonians and calibrated charge operator

We couple one $h_{11/2}$ neutron quasiparticle to an s, d boson core. The boson Hamiltonian is

$$H_B = \epsilon_d n_d + \kappa Q_\chi \cdot Q_\chi, \tag{1}$$

$$Q_\chi = s^\dagger \widetilde{d} + d^\dagger s + \chi [d^\dagger \times \widetilde{d}]^{(2)}. \tag{2}$$

The ordinary product $Q_\chi \cdot Q_\chi$ includes its one-body contractions. The boson–fermion interaction contains the quadrupole, normal-ordered exchange and monopole terms of Ref. [5], restricted to the single $j = 11/2$ shell. With the Condon–Shortley convention,

$$\gamma_{jj} = (1 - 2v^2) Q_{jj}, \quad \beta_{jj} = 2\sqrt{v^2(1-v^2)}\, Q_{jj}, \tag{3}$$

$$\Gamma_{jj} = \Gamma_0 \gamma_{jj}, \qquad \Lambda^j_{jj} = -2\Lambda_0 \sqrt{\frac{5}{2j+1}}\, \beta^2_{jj}. \tag{4}$$

Here $Q_{jj} = \langle h_{11/2} \| Y_2 \| h_{11/2} \rangle = -\sqrt{525/(143\pi)}$ is an angular matrix element; no extra radial factor is inserted. The scalar monopole term is $A_0 n_d$ for one fermion. The single-shell fermion energy supplies only a common offset.

Table 1 gives the fixed parameters from Tables I, III and V of Ref. [5]. We associate each odd isotope with its $A - 1$ core and use $N_B = 8, 7, 6, 5, 4$. This assignment follows the stated core chain and conventional proton-particle/neutron-hole pair counting. The original author input cards are unavailable, so it remains a declared reconstruction.

For ^{135,137}Ba, we independently extracted all six negative-parity theory series displayed in the archived arXiv-v2 Fig. 5(c) of Ref. [5] before comparing the fixed calculation. With the lowest $11/2$ state as the common zero, the ordering agrees and the largest nonreference differences are 1.896 and 3.128 keV, respectively. These resolved differences are not experimental uncertainties. The figure contains no $19/2$ or higher same-spin states; this comparison does not recover author wavefunctions or certify the isomer mapping used below.

The primary electric-quadrupole tensor is

$$T(E2) = e_B Q_{\chi'} - \frac{e_F (1 - 2v^2) Q_{jj}}{\sqrt{5}} [a^\dagger \times \widetilde{a}]^{(2)}, \tag{5}$$

with $\chi' = 0.35$ and $e_F = 0.125\, e\,\mathrm{b}$, as specified in Ref. [5]. Each e_B is calibrated to one evaluated first-2^+ core strength [6]. The downward $B(E2; 2^+_1 \to 0^+_1)$ values for ^{128,130,132,134,136}Ba are respectively 0.276525, 0.2326, 0.1720, 0.1358 and $0.0814\, e^2\mathrm{b}^2$. The solver uses the upward values, five times larger. For ^{128}Ba, the selected value is derived from $T_{1/2} = 105(9)$ ps, $E_\gamma = 284.10$ keV and total conversion coefficient 0.0539, using $\lambda_\gamma = 1.2239 \times 10^{13} E_\gamma^5 B(E2)$ with MeV and $e^2\mathrm{b}^2$ units. The other selections are evaluated absolute Coulomb-excitation strengths. Evaluation versions, alternatives, source locators and uncertainty dependencies are supplied with the data.

The lowest $11/2^-$ state is selected in each complete fixed-$M = 1/2$ block. A core-only control sets $e_F = 0$ without changing the state or e_B. This distinguishes operator effects from a change of wavefunction.

2.2 Complete and selected-channel moments

Write $R_{fi} = \langle J_f \alpha_f \| T(E2) \| J_i \alpha_i \rangle$ and use

$$\langle J_f M_f | T_{2\mu} | J_i M_i \rangle = \frac{C(J_i M_i, 2\mu \mid J_f M_f)}{\sqrt{2J_f + 1}} R_{fi}. \tag{6}$$

The complete quadratic and signed cubic moments are

$$q_2 = \langle T \cdot T \rangle = \frac{1}{2J+1} \sum_a |R_{ai}|^2, \tag{7}$$

Table 1 Fixed Hamiltonian inputs and the positive boson charge used in the primary calculation. Energy parameters are in MeV and e_B in $e\,\mathrm{b}$. The occupation v^2 is a supplied mean-field probability, not a measured spectroscopic factor. No parameters were fitted to the complete–restricted differences.

Odd Ba	Core Ba	N_B	v^2	ϵ_d	κ	χ	Γ_0	Λ_0	A_0	e_B
129	128	8	0.453	0.03	−0.102	−0.18	0.6	2.1	−0.15	0.132680
131	130	7	0.568	0.06	−0.116	−0.18	0.6	2.1	−0.23	0.133741
133	132	6	0.687	0.13	−0.122	−0.18	0.6	0.9	0	0.127686
135	134	5	0.810	0.38	−0.124	−0.24	0.6	1.0	−0.90	0.129842
137	136	4	0.936	1.15	−0.122	−0.85	0.4	5.0	−0.60	0.125640

$$q_3 = -\sqrt{\frac{7}{2}}\langle [T\times T]^{(2)}\cdot T\rangle = -\frac{(-1)^{2J}}{2J+1}\sqrt{\frac{35}{2}}\sum_{ab}\begin{Bmatrix} 2 & 2 & 2\\ J & J_a & J_b\end{Bmatrix} R_{ia}R_{ab}R_{bi}. \tag{8}$$

These are standard scalar-closure relations [1, 3, 7]. "Complete" here means all contributing states within the declared finite model space. For $J = 11/2$, the intermediate spins are 7/2, 9/2, 11/2, 13/2 and 15/2. Diagonal matrix elements and transitions to other states of spin 11/2 must both be included.

Hermitian reciprocity is $R_{ab} = (-1)^{J_a-J_b}R^*_{ba}$. The order of tensor operators in Eq. (8) is retained; components need not commute. The full single-j interaction and tensor conventions are specified in Appendix A.

We report the signed ratio $r = q_3/q_2^{3/2}$. When $|r| \le 1$, the folded rigid-equivalent coordinate is $\bar{\gamma}_{\mathrm{mom}} = \frac{1}{3}\arccos|r|$. This conversion alone does not determine the rigidity or fluctuations of an intrinsic shape. Neither projected operators nor fluctuating charge distributions guarantee that every normalized third moment is in the real arccos domain.

For comparison, we implement the selected-channel prescription of Eqs. (12)–(15) of Ref. [5]. With a, b restricted to the lowest n states of $L = J + 2$, define

$$S_n = \sum_{a\in L,n}|R_{ai}|^2, \qquad C_n = \sum_{a,b\in L,n} R_{ia}R_{ab}R_{bi}, \tag{9}$$

$$z_n = \sqrt{\frac{7}{10}}\frac{|C_n|}{S_n^{3/2}}, \qquad \gamma_{\mathrm{sel}}(n) = \frac{1}{3}\arccos z_n. \tag{10}$$

The selected multiplets are ordered by excitation energy. Their projectors commute with rotations, and state phases cancel around the closed products. Thus Eq. (10) remains rotationally invariant. It differs from Eq. (8) through the restriction, recoupling coefficients and normalization. We retain z_n unchanged and leave γ_{sel} undefined when $z_n > 1$.

Two fractions separate numerical saturation from coverage:

$$c_n = \frac{S_n}{S_\infty}, \qquad f_n = \frac{S_n}{(2J+1)q_2}. \tag{11}$$

c_n measures convergence within $J + 2$, whereas f_n measures its contribution to the complete quadratic moment.

2.3 Numerical controls

The normalized boson occupation basis and a single fermion define the full space; only fixed-M blocks are diagonalized densely. At $N_B = 8$, the product space has dimension 15444 and its $M = 1/2$ block has dimension 965. Degenerate energy eigenspaces are resolved with J^2, and the Hamiltonian, angular momentum and orthonormality residuals are checked.

For every isotope and both charge operators, direct contraction of the five laboratory tensor components is compared with Eqs. (7)–(8). In the five primary cases, the largest absolute quadratic and cubic discrepancies are $4.6\times10^{-13}\,e^2\mathrm{b}^2$ and $8.5\times10^{-13}\,e^3\mathrm{b}^3$. Independent analytic controls include a spectator $R_{\mathrm{core}} = 0\otimes j$ state, for which the core-only complete moments equal the core moments. Its $J + 2$ fraction is already $(2j + 5)/[5(2j + 1)] = 4/15$ for $j = 11/2$; incomplete coverage by that channel alone is therefore not an interacting-system anomaly.

3 Results

3.1 A control with exactly fixed charge geometry

To separate intrinsic geometry from angular coupling, we constructed an independent unpaired particle–rotor control with $j = 11/2$, fixed triaxiality $\gamma = 20°$ and a core-only electric tensor

$$T_{2\mu}(\Omega) = \mathcal{Q} \sum_{\nu} D^{2*}_{\mu\nu}(\Omega) a_{\nu}, \qquad (12)$$

$$a_0 = \cos\gamma, \qquad a_{\pm 2} = \frac{\sin\gamma}{\sqrt{2}}.$$

The remaining intrinsic components vanish. Because these are multiplication operators on rotor orientation, the standard tensor contractions give the pointwise identities

$$T \cdot T = \mathcal{Q}^2 \mathbf{1}, \qquad -\sqrt{\frac{7}{2}} [T \times T]^{(2)} \cdot T = \mathcal{Q}^3 \cos 3\gamma \, \mathbf{1}. \qquad (13)$$

They hold independently of the particle–rotor eigenstate. We set $\mathcal{Q} = 1$ and quote moments in the corresponding amplitude units. The complete ratio is therefore exactly $1/2$ and its angle $20°$.

The Hamiltonian retains fixed irrotational inertias while the coefficient C of a declared quadrupole single-particle field takes the preselected values 0, 1, 2, 4 and 8 MeV. The field changes angular coupling but neither the tensor in Eq. (12) nor its geometry. Appendix C specifies the field normalization, body-axis convention and physical D_2 space. This is an illustrative control, with no Ba fit and no identification of C with the IBFM's Γ_0.

Figure 1 shows that the all-15/2 selected ratio increases from 0.32262 to 1.02625 while both complete moments remain fixed. All 48 physical 15/2 states are included, so the final domain failure survives removing the state-count restriction within that spin. The corresponding static fraction grows from zero to 35.95%. Thus a change in angular correlations can alter this converged diagnostic without changing intrinsic charge shape. This result does not establish unchanged geometry in the separate IBFM calculation.

Table 2 Primary model results. q_2 is in $e^2\mathrm{b}^2$; fractions are defined in Eq. (11). The signed r and absolute-chain z_∞ use different contractions. $z_\infty > 1$ has no real selected angle.

A	q_2	$100c_5$	$100f_5$	r	z_∞
129	1.53187	99.62	28.90	+0.0055	0.0351
131	1.26392	99.59	27.56	−0.0700	0.1334
133	0.94637	99.84	24.31	−0.1447	0.6369
135	0.67662	99.85	20.00	−0.0505	1.0241
137	0.47677	99.62	21.50	−0.0667	0.7382

The charge identities were checked as matrices on the entire initial 11/2 space, with maximum residual 5.6×10^{-16}, and against independent Euler-angle integration of tensor elements. All required intermediate spins for its quadratic and cubic closure are present. Numerical branch matching retains the lowest 11/2 state in both directions, with minimum adjacent squared overlap 0.648. The exact geometric identity does not depend on extrapolating that sampled tracking into a continuous no-crossing claim.

3.2 Channel saturation and total charge strength

We next examine the same diagnostic in the IBFM family, which does not impose the rigid-charge operator identities. Table 2 shows that complete charge strength decreases across the chain, from $q_2 = 1.53187$ in ^{129}Ba to $0.47677\, e^2\mathrm{b}^2$ in ^{137}Ba. The ratio is 0.3112. The comparatively weak evolution of the complete normalized cubic moment therefore cannot be interpreted as constant collectivity.

The first five $15/2^-$ states contain 99.59–99.85% of the quadratic strength within that spin, but only 20.00–28.90% of complete q_2 (Table 2). Explicit evaluation with every $15/2^-$ state changes the cubic ratio by $|z_5 - z_\infty| < 0.0093$. For the small ^{129}Ba ratio this is a 10.7% deviation relative to z_∞, despite an angle shift of only 0.0715°. In the four cases with a real selected angle, its change is at most 0.259°. These raw comparisons test cubic convergence directly instead of inferring it from quadratic saturation.

The largest qualitative separation occurs in ^{135}Ba. Its complete signed ratio is −0.05052, whereas $z_5 = 1.02611$. Including the entire

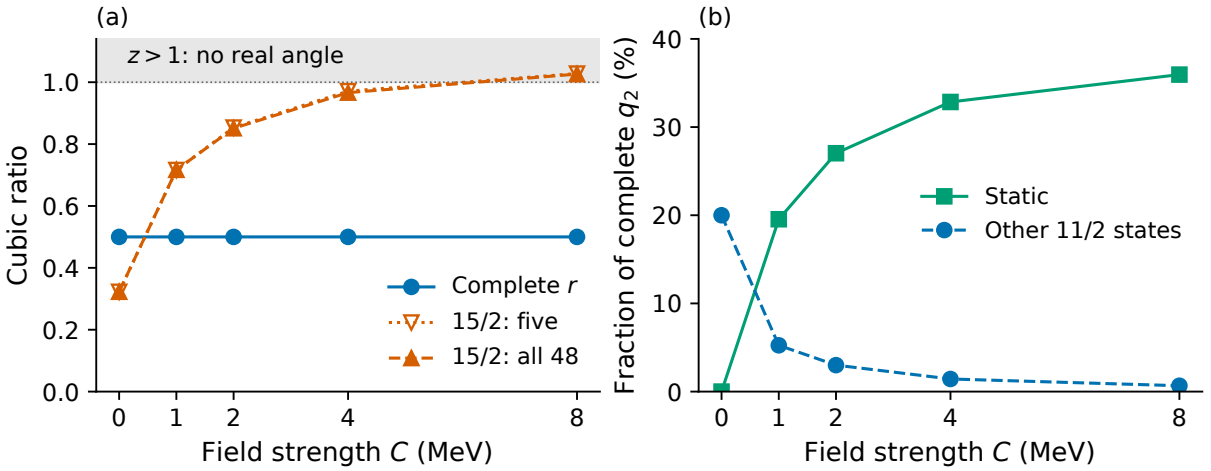


Fig. 1 Independent rigid-charge particle–rotor control at $\gamma = 20°$. (a) Complete ratio and selected-channel ratios using five or all 48 15/2 states. (b) Static and other-same-spin fractions of q_2. The intrinsic tensor and inertias remain fixed at every point; C has the normalization of Appendix C. The shaded region marks the absence of a real selected angle. Lines join the five calculated points and do not assert untested continuous-parameter behavior.

selected channel yields $z_\infty = 1.02414$, so its failure to define a real angle persists after within-channel completion. In the same state with only the core part of the charge operator, the five-state angle is 3.749° and the complete folded angle is 28.561°. The fermion term changes an already near-boundary selected coordinate into an out-of-domain ratio. The central distinction is therefore visible before the domain crossing; it does not depend on interpreting the crossing as a physical instability.

3.3 Static strength and signed cubic cancellation

The quadratic decomposition in Fig. 2(a) reveals a large change in the same-spin sector. In ^{135}Ba, the static diagonal term carries 33.69% of complete q_2, and transitions to other $11/2^-$ states carry another 3.74%. The corresponding static fractions in ^{129}Ba and ^{131}Ba are only 0.287% and 0.264%. Thus the enhanced same-spin strength is primarily static in the ^{135}Ba case, while the same-spin channel in the lighter pair is dominated by transitions to other states. A description based solely on the selected $15/2^-$ strength does not retain this redistribution.

The complete cubic moment is small because substantial positive and negative contributions cancel. For ^{135}Ba, their spin-sector sums are $+0.29086$ and $-0.31897\,e^3\mathrm{b}^3$, respectively, giving $q_3 = -0.02812\,e^3\mathrm{b}^3$. The 15/2–15/2 sector alone contributes $-0.06382\,e^3\mathrm{b}^3$. It is opposed by the 11/2–11/2 sector, $+0.12335\,e^3\mathrm{b}^3$, and by both 11/2–13/2 sectors, whose combined contribution is $+0.11950\,e^3\mathrm{b}^3$. Other sectors complete the cancellation shown in Fig. 2(b).

Equation (10) retains only the 15/2–15/2 path, with its own normalization and coefficient. Adding more states within that path cannot recover interference with the omitted spins. The different cubic trends thus correspond to different combinations of the same matrix-element network. On the common denominator $q_2^{3/2}$, the selected spin sector contributes -0.11467 and all remaining sectors contribute $+0.06415$, producing $r = -0.05052$. This decomposition keeps the normalization fixed when comparing the retained and omitted paths.

The operator decomposition also matters after this cancellation. In the fixed ^{135}Ba state, the core-only cubic moment is $-0.03981\,e^3\mathrm{b}^3$. Terms containing one or more fermion tensors add $+0.01170\,e^3\mathrm{b}^3$, moving the complete value toward zero. A small fermion contribution to quadratic strength is therefore insufficient grounds to neglect it in a normalized cubic diagnostic.

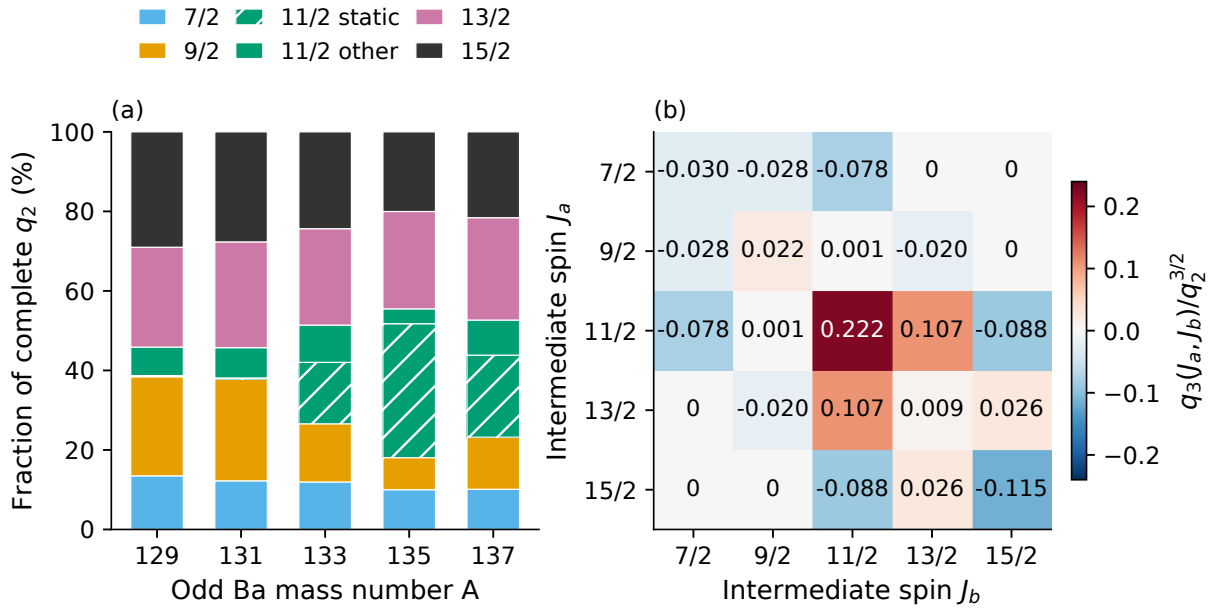


Fig. 2 Decomposition of complete charge moments. (a) Contributions to q_2 by intermediate spin. The hatched $11/2$ segment is the static matrix element; the solid segment contains other states of the same spin. (b) Signed contributions to $q_3/q_2^{3/2}$ in ^{135}Ba, grouped by the ordered pair of intermediate spins. Both matrix orientations are retained in the total sum. Exact angular-momentum selection zeros are shown as zero; other labels are rounded for display.

3.4 A controlled quadrupole-coupling path

To isolate a dynamical origin, we varied the ^{135}Ba quadrupole coupling as $\Gamma_0 = 0.6s$ MeV, with $s = 0, 0.25, 0.50, 0.75, 1$, while keeping the exchange and monopole strengths, core, occupation and charge operator fixed. These are interventions in the model, not alternative fits to the nucleus. Forward and backward maximum-overlap tracking both select the lowest $11/2$ state throughout the sampled path. Adjacent squared overlaps exceed 0.907, and the smallest sampled gap to the next state of the same spin is 0.481 MeV. Thus no lowest-state switch is observed at this resolution.

Figure 3 shows a strong redistribution with little change in total strength. Complete q_2 remains between 0.67662 and 0.69413 $e^2\mathrm{b}^2$, a range of less than 2.6%, and $\langle n_d \rangle$ stays between 2.324 and 2.354. The static fraction, however, rises from 0.0091% to 33.69%. The core-$R = 2$ probability increases from 0.0151 to 0.4837, while the core-$R = 0$ probability falls from 0.9522 to 0.4886. In this controlled comparison, quadrupole coupling principally changes angular recoupling and static polarization rather than the amount of quadratic charge strength.

The selected ratio rises through a specific change of its numerator and denominator. Over the path, S_∞ falls from 2.32070 to 1.62594 $e^2\mathrm{b}^2$, whereas the signed chain C_∞ grows from 0.37162 to 2.53787 $e^3\mathrm{b}^3$. In the intermediate energy eigenbasis, its middle-diagonal contribution changes from -0.73467 to $+1.16709\, e^3\mathrm{b}^3$: cancellation with off-diagonal paths is replaced by reinforcement. Consequently z_∞ rises from 0.08795 to 1.02414. At the final point, the additional spin sectors in the complete cubic moment still largely cancel. Since the all-15/2 projector is unchanged along this path, the result is a change of the initial state's charge response, not a consequence of reordering a five-state list.

3.5 Charge-operator and static-moment constraints

The distinction between channel saturation and full coverage survives 35 operator/calibration settings, evaluated on the fixed Ba states: the primary setting, two fermion charges, two alternative χ' values and both quoted-margin core-strength

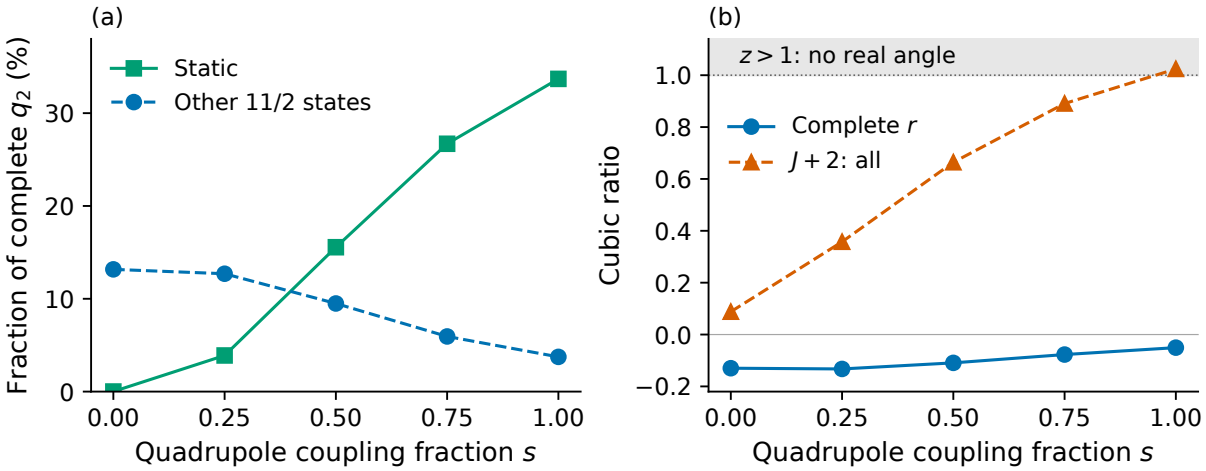


Fig. 3 Fixed-exchange/monopole control for ^{135}Ba: $\Gamma_0 = 0.6s$ MeV, $\Lambda_0 = 1$ MeV, $A_0 = -0.9$ MeV. The core and electric-quadrupole operator are unchanged. (a) Static and other-same-spin fractions of complete q_2. (b) Complete signed ratio and the all-15/2 selected ratio. Points are the five calculated settings; lines guide the eye. The shaded region has no real selected angle.

endpoints for each isotope. Across these settings, the full 15/2 channel carries 14.12–29.24% of q_2, and its first five states exhaust 99.588–99.939% of that channel. These are deterministic sensitivity tests, not a parameter posterior.

The actual cubic coordinates are more sensitive. In ^{135}Ba, $\chi' = 0$ or $\chi' = \chi = -0.24$ removes the selected-domain failure and changes the complete cubic sign. The complete/selected folded angles become 24.418°/6.241° and 20.166°/10.659°, respectively. Changing an E2 operator on one wavefunction is not a change of Hamiltonian shape.

Static moments further constrain nuclear interpretation. The primary ^{137}Ba value, $Q_s = +0.77296\,e\,\mathrm{b}$, agrees with its source-named calibration target $+0.78(9)\,e\,\mathrm{b}$ and is not an independent test. The ^{135}Ba prediction, $+1.17693\,e\,\mathrm{b}$, exceeds the evaluated $+0.96(8)\,e\,\mathrm{b}$; Ref. [5] separately quotes +0.98(8) experimentally and $+1.13\,e\,\mathrm{b}$ theoretically. The source variants are not averaged. The primary ^{136}Ba core moment also disagrees with several evaluated calibration branches. These discrepancies limit the quantitative static redistribution that can be assigned to the actual nucleus.

We retained all five finite central-value operator calibrations obtained from the distinct ^{136}Ba moment targets, the common core strength and $Q_s(^{137}\mathrm{Ba}) = 0.78\,e\,\mathrm{b}$, without using ^{135}Ba to select a branch. Two branches improve its predicted moment to 0.923 and $0.915\,e\,\mathrm{b}$ but give complete ratios $r = -1.093$ and -1.096. Thus even complete moments need not admit a real rigid-equivalent angle. Appendix D retains every branch, the shared-source and unknown-covariance limitations, and a marginal truncation case. Matching Q_s through the coupling instead changes the spectrum (Appendix B); neither calibration is an independent validation of the electromagnetic network.

4 Interpretation and scope

The fixed-spin comparison separates two structural questions: how much charge-quadrupole strength a state carries, and how particle–core coupling distributes that strength and its interference paths among angular momenta. A selected-channel diagnostic can be sensitive to the second question even after the first is only partially represented. Its sensitivity is potentially useful. Selected matrix-element ratios can also give deformation estimates consistent with invariant analyses within specified low-lying even-even collective regimes [8]. Applying a conversion to a mixed odd-A state requires a corresponding test of its coupling regime and charge response.

Prior work establishes approximations to quadrupole moments [2, 3], coupling-dependent ΔJ indicators in odd nuclei [4], and differences between deformation definitions near $A = 130$ [7].

Microscopic restricted-network studies also show how configuration mixing and coverage affect inferred shapes [9]. The present mechanism test combines a fixed-operator intervention with a geometrically fixed control. The former resolves how static strength and signed paths respond in the declared Ba family; the latter demonstrates that variation of the selected diagnostic does not require a change of intrinsic charge shape. Neither the invariant identities nor the existence of channel dependence is new.

The rigid-charge identity is not a claim that the IBFM wavefunctions have unchanged geometry. Their finite $s, d \otimes h_{11/2}$ space omits configurations, and products of its effective tensor need not reproduce the full nucleus's charge moments. The operator alternatives, static-moment discrepancy and conditional isomer failure limit empirical interpretation. Neither a moment angle near 30° nor a domain crossing identifies γ softness, rigidity or a collective mode. Furthermore, the published invariant figures of Ref. [5] remain unreproduced. A difference is already present for its even-core $J = 0$ case, where the spin restriction cannot explain it. Our conclusions concern the specified operators and states.

4.1 Empirical reach of the Ba reconstruction

A separate high-spin test limits the spectroscopic application. The tentative ^{137}Ba $(19/2^-)$ isomer has a reported 589(20)-ns half-life and $B(E2; 19/2 \to 15/2) = 0.46(3)$ W.u. for its 120.1-keV decay [10]. The strength is already corrected for conversion and derives from the same decay evidence as the lifetime. For each of the two tested model parents, all energetically open E2 photon rates give

$$T_{1/2} \leq \frac{\ln 2}{\sum_f \lambda_{\gamma,f}(E2)}. \tag{14}$$

Across the primary, core-only and five conditional operators, the largest upper bound is 2.811×10^{-4} ns. The unchanged modeled schemes cannot represent the observed isomer under these state identifications. This bound uses model energies and amplitudes; additional conversion or other positive rates can only shorten the lifetime of the same scheme.

An alternate selected branch approaches the observed strength, but a dominant fast route remains open (Appendix E). The source-spectrum comparison contains no 19/2 state and cannot certify that mapping or attribute the failure to the original authors. This test limits the reconstruction's high-spin reach without directly testing the initial-11/2 cubic loop, whose rank-two intermediate spins end at 15/2.

An observable consequence can be tested before determining the full cubic network. Define the static-to-first-transition ratio

$$R^{(1)}_{\rm stat} = \frac{D_0}{B(E2; 11/2^-_1 \to 15/2^-_1)}, \qquad D_0 = \frac{5}{16\pi}\frac{Q_s^2}{55/91}. \tag{15}$$

Here $55/91 = C(11/2, 11/2; 2, 0 \mid 11/2, 11/2)^2$ and D_0 is the diagonal contribution to q_2, not a decay strength. The primary model predicts $R^{(1)}_{\rm stat} = 2.2722$ and 1.3565 for ^{135,137}Ba, respectively. For a measured reverse decay, $B_\uparrow = (16/12)B_\downarrow$. This first-state ratio differs from a sum over all 15/2 states; the first state carries about 74% of that model channel in ^{135}Ba. An experimental test requires the corresponding level assignment, static moment and absolute E2 strength. Combining an experimental numerator with a model denominator would not produce an experimental ratio. Signed intermediate-state matrix elements would still be needed to test the cubic cancellation itself.

The audited records provide candidate $(15/2^-)$ levels at 950.5 keV in ^{135}Ba and 2229.8 keV in ^{137}Ba, with respective 682.3- and 1568.2-keV connections to the $11/2^-$ isomers [6, 10]. They do not supply the corresponding absolute E2 denominators. The upper ^{137}Ba 19/2 lifetime used in the separate check of Sec. 4.1 cannot replace the 15/2 parent lifetime. A determination of this first-transition strength, together with the static moment, would directly test the predicted redistribution before attempting the full cubic network. This limitation concerns the specified source set; the tentative assignments are retained.

5 Conclusions

Selected-channel convergence does not establish an intrinsic-shape interpretation for a mixed odd-A state. In the declared Ba model, quadrupole coupling redistributes static strength and selected cubic paths while complete quadratic strength changes little. An independent rigid-charge rotor demonstrates the same diagnostic sensitivity with geometry exactly fixed. Static-moment discrepancies, operator alternatives and the conditional high-spin isomer failure bound the nuclear application. Reporting raw moments, angular-channel coverage and signed contributions alongside state-count convergence makes these distinct questions testable.

A Single-shell interaction and tensor ordering

With $F^{(k)} = [a_j^\dagger \times \widetilde{a}_j]^{(k)}$, the interaction used in the IBFM calculations is

$$\begin{aligned} H_{BF} &= \Gamma_{jj} Q_\chi \cdot F^{(2)} + \Lambda^j_{jj} X \\ &\quad - A_0 \sqrt{2j+1}\, n_d F^{(0)}, \\ X &= :\Big[[d^\dagger \times \widetilde{a}_j]^{(j)} \times [a_j^\dagger \times \widetilde{d}]^{(j)}\Big]^{(0)}: . \end{aligned} \tag{16}$$

The tilde conventions are $\widetilde{a}_{jm} = (-1)^{j-m} a_{j,-m}$ and $\widetilde{d}_\mu = (-1)^{2-\mu} d_{-\mu}$. The exchange term is explicitly normal ordered, retaining the fermion interchange sign and omitting contractions. Since $F^{(0)} = -1/\sqrt{2j+1}$ in the one-fermion space, the last term is $A_0 n_d$. Changing it in a control therefore changes the odd-system n_d coefficient, even when the bare even-core Hamiltonian is fixed.

Scalar and coupled products follow

$$A^{(k)} \cdot B^{(k)} = \sum_\mu (-1)^\mu A_\mu B_{-\mu}, \tag{17}$$

$$[T \times T]^{(2)}_\nu = \sum_{\mu\eta} C(2\mu, 2\eta \mid 2\nu) T_\mu T_\eta. \tag{18}$$

Operator order and a common eigenvector phase convention are preserved. No state-independent magnitudes are substituted for signed closed products.

B Static-moment matching along the coupling path

Along the fixed-operator path of Fig. 3, matching the preselected ^{135}Ba values $Q_s = 0.88, 0.96, 1.04\, e\,\mathrm{b}$ gives $\Gamma_0 = 0.33378, 0.37929, 0.43683$ MeV. The roots were fixed using only signed static moments before evaluating the new scalar diagnostics. The complete folded angles are $28.050°, 28.240°, 28.473°$, whereas the selected-all angles are $14.520°, 12.371°, 9.647°$. The separation remains $13.53°$–$18.83°$, with both angles real. All four points, including the separately retained paper-quoted $Q_s = 0.98\, e\,\mathrm{b}$ alternative, pass direct/reduced closure and bidirectional matching to the neighboring sampled states.

This is conditional calibration, not an independent validation or a simultaneous spectroscopic fit. The central match moves the calculated $15/2^-_1 - 11/2^-_1$ spacing from 691.9 to 526.6 keV, while the tentative experimental connection is 682.3 keV. For the central $Q_s = 0.96\, e\,\mathrm{b}$ match, the largest absolute change in the first two states of each spin from 7/2 to 19/2, referenced to the corresponding lowest 11/2 state, is 0.3684 MeV. Across all four matched points, that maximum is 0.4322 MeV, at the $Q_s = 0.88\, e\,\mathrm{b}$ target. The static moment has been consumed as a calibration input and its agreement cannot be counted again as independent evidence.

C Rigid-charge particle–rotor control

Using standard particle–rotor kinematics [11], we assemble the unpaired Hamiltonian directly:

$$H(C) = \sum_{k=1}^{3} A_k R_k^2 + C\, 1 \otimes Q_j(\gamma), \tag{19}$$

$$\begin{aligned} R_k &= I_{k,\mathrm{right}} \otimes 1 - 1 \otimes j_k, \\ A_k &= \frac{\hbar^2}{2\mathcal{J}_k}, \quad \mathcal{J}_k = \frac{4}{3}\mathcal{J}_0 \sin^2(\gamma - 2\pi k/3). \end{aligned} \tag{20}$$

$$\begin{aligned} Q_j(\gamma) &= \frac{\cos\gamma}{4j(j+1)}[3j_z^2 - j(j+1)1] \\ &\quad + \frac{\sqrt{3}\sin\gamma}{4j(j+1)}(j_x^2 - j_y^2). \end{aligned} \tag{21}$$

The angular momentum components are dimensionless, in units of $\hbar$; $\mathcal{J}_0 = 30\,\hbar^2/\mathrm{MeV}$, $j = 11/2$ and $\gamma = 20°$. The unsorted principal inertias are 38.79385, 16.52704 and 4.67911 $\hbar^2/\mathrm{MeV}$. The field Q_j is dimensionless; its normalization defines C and is distinct from the core-only electric tensor. There is no pairing projection or zero-gap approximation.

The right-body representation is $I_{\mathrm{right}} = (J_x, -J_y, J_z)$, with $[I_1, I_2] = -iI_3$; the particle matrices have ordinary commutators. The full physical D_2 subspace is retained at each $I = 1/2, \ldots, 15/2$. These 216 multiplets exhaust all required 7/2–15/2 intermediate states for initial $I = 11/2$ quadratic and cubic closure. They are not a globally complete infinite-rotor space: for an initial $I = 15/2$, additional 17/2 and 19/2 channels would be required.

Equation (13) follows from rotation invariance of the intrinsic tensor polynomials. Equivalently, its real traceless Cartesian tensor has $\mathrm{tr}\,A^2 = 1$ and $\sqrt{6}\ \mathrm{tr}\,A^3 = \cos 3\gamma$ for $\mathcal{Q} = 1$. The numerical contractions equal these constants times the identity on the 36-dimensional initial-$I = 11/2, M = 1/2$ coordinate space. Independent Euler-angle integration agrees with the product-basis matrix elements within 2.8×10^{-14}. The independent D_2 group projector, tensor covariance and Hermiticity are checked as well.

At $C = 0$, the exact $R_{\mathrm{core}} = 0$ spectator has zero rotational energy and static moment. Its spin fractions equal $(2I_f + 1)/[5(2j+1)]$. The two core-$R = 2$ energies, 0.126795 and 0.473205 MeV, are recovered independently from a five-dimensional integer-spin rotor. A deliberately incorrect body-axis sign violates the zero-energy spectator condition, and omission of the 15/2 outer sector loses 4/15 of the initial quadratic closure. These checks test different parts of the construction without fitting the selected ratio.

For a fixed axial $K = J$ rotor with intrinsic tensor amplitude $\mathcal{Q} = T^{\mathrm{body}}_{20}$, the complete moments are $q_2 = \mathcal{Q}^2$ and $q_3 = \mathcal{Q}^3$. The conventional electric intrinsic quadrupole moment instead satisfies $\mathcal{Q} = \sqrt{5/(16\pi)}Q_0$. This control illustrates the regime dependence of Eq. (10): at $J = 11/2$ it gives 10.193° despite axial input. These controls validate the definitions and implementation; they are not new shape theorems or empirical uncertainty estimates.

Table 3 All conditional moment-calibration branches at fixed Hamiltonians. t_{136}, e_F and $Q_s(135)$ are in $e\,\mathrm{b}$. Ratios refer to the lowest ^{135}Ba $11/2^-$ state. Both $|r| > 1$ and $z_\infty > 1$ imply an undefined real angle.

t_{136}	χ'	e_F	$Q_s(135)$	r	z_∞
−0.19	0.982	0.199	1.089	−0.642	1.030
+0.01	1.959	0.311	0.923	−1.093	0.850
+0.02	2.004	0.317	0.915	−1.096	0.840
+0.07	2.222	0.344	0.873	−1.092	0.792
+0.25	2.943	0.445	0.723	−0.920	0.640

D Conditional charge calibrations

For each distinct central ^{136}Ba 2_1^+ moment target, the boson charge is kept positive and recalibrated to the common $B_\uparrow = 0.407\,e^2\mathrm{b}^2$. Its moment determines χ', and $Q_s(^{137}\mathrm{Ba}) = 0.78\,e\,\mathrm{b}$ then determines e_F. All finite sign-consistent roots are retained; the ^{135}Ba moment is not used. Its boson charge matches the unchanged ^{134}Ba core strength.

The independent E2 parameter χ' is not restricted here by a customary interval for the Hamiltonian parameter χ. These are central-value comparisons at a common averaged core strength: moment and strength branches may covary, and the 135/137Ba moment evaluations share source lineage. No joint confidence region or preference for a branch is inferred.

A marginal domain crossing illustrates the separate role of state-count truncation. For ^{137}Ba with the $+0.01\,e\,\mathrm{b}$ core target, $z_5 = 1.00027$ but $z_\infty = 0.99484$, so completing that spin channel restores its real selected angle. The complete ratio remains $r = -1.08957$. This differs from the primary ^{135}Ba crossing that persists after within-spin completion.

E Isomer units, mappings and decay completeness

The experimental parent and daughter are the tentative $(19/2^-)$ 2349.9-keV and $(15/2^-)$ 2229.8-keV levels, respectively. Their common anchor is the 661.7-keV $11/2^-$ isomer. For every open model branch, the photon rate used is

$$\lambda_{\gamma,f}(E2) = 1.2239 \times 10^{13} E_{if}^5 B(E2; i \to f), \quad (22)$$

Table 4 All four predeclared state pairs for the primary ^{137}Ba operator. Relative energies use the lowest model 11/2 state as zero. The experimental candidate energies on the same reference are 1688.2 and 1568.1 keV, respectively. Model-state ordinals do not establish experimental identities.

$19/2_i \to 15/2_f$	E_i (keV)	E_f (keV)	$B_\downarrow$ (W.u.)
$1 \to 1$	2142.484	968.874	22.46885
$1 \to 2$	2142.484	2094.159	2.17762
$2 \to 1$	3648.848	968.874	0.05585
$2 \to 2$	3648.848	2094.159	13.14225

with energy, strength and rate units MeV, $e^2\mathrm{b}^2$ and s^{-1}. No unobserved modeled daughter is treated as an experimentally excluded level.

The ordinary downward Weisskopf reference is

$$B_{\mathrm{W}}(E2) = \frac{1}{4\pi}\left(\frac{3}{5}\right)^2 (1.2A^{1/3})^4\, e^2\mathrm{fm}^4. \qquad (23)$$

For $A = 137$, it equals $0.00419544833\, e^2\mathrm{b}^2$. Thus the reported 0.46(3) W.u. corresponds to $(1.930 \pm 0.126) \times 10^{-3}\, e^2\mathrm{b}^2$, with only the source's quoted margin converted. Our downward strength is $|R_{15/2,19/2}|^2/20$, while the reverse strength is larger by 20/16. This is different from the 16/12 factor in Eq. (15) and the factor five for a $0 \leftrightarrow 2$ core transition.

All seven operators—the primary, core-only and five conditional calibrations—were evaluated for every pair. The energy-distance mapping was fixed before the E2 evaluation. The nearest of the four pairs is $19/2_1 \to 15/2_2$, whose parent and daughter residuals are still +454 and +526 keV. This minimum distance is not a certified physical correspondence or a statistical chi-square fit. The all-open-width test was defined afterward as a separate follow-up, without refitting either Hamiltonian or operator. The full package retains 28 strengths, 14 parent/operator sums and 84 open E2 branches.

For the primary operator, the two parent half-life upper bounds are 2.69817×10^{-4} and 6.02153×10^{-5} ns. With the conditional $t_{136} = +0.07\, e\,\mathrm{b}$ operator, the alternative $1 \to 2$ strength is 0.42002 W.u., outside the quoted interval $[0.43, 0.49]$ W.u. Its modeled photon energy is only 48.325 keV. The competing $1 \to 1$ branch has 21.86655 W.u. and accounts for 99.9981% of that model's E2 photon rate. These are model rate fractions, not measured total branching probabilities.

No experimental photon energy is substituted into a different model branch. Missing conversion and non-E2 rates are not assumed zero to claim a complete lifetime prediction. Their nonnegativity gives the conditional upper bound in Eq. (14); it cannot compensate for the fast E2 routes of the unchanged model scheme.

Data and code availability. The accompanying `data/` directory contains the selected evaluated core strengths with provenance, fixed Hamiltonian inputs, calculated scalar decompositions and Python code. The additional rigid-charge, isomer and source-spectrum checks are packaged separately in `data/revision-v2/`; their producers, adaptation boundaries and replay instructions accompany the results. The original 45-file manifest is preserved. The code requires NumPy and SciPy; figure scripts also use Matplotlib. No private experimental data or publisher full-text files are included. A permanent public repository identifier has not yet been assigned.

AI assistance and draft status. AI assistance was used for literature retrieval, source comparison, algebraic and numerical implementation, verification and manuscript drafting. The calculations and source records are supplied for inspection. Human author information, contributions and final accountability statements remain to be provided; this manuscript has not been submitted.

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